



Indian Association for the Cultivation of Science

(Deemed to be University under *de novo* Category)

Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's Program/PhD Course
Mid-Semester Examination-Autumn 2025

Subject: Molecules of life and cells
Full Marks: 25

Subject Code(s): BIS1101
Time Allotted: 2 h

Answer Part A and part B separately
PartA (Answer all)

Q1 1 mark for each correct answer

(5 marks)

- a) The process of copying a gene's DNA sequence into a sequence of RNA is called
A) Replication. B) Transcription. C) Translation. D) PCR
- b) Which molecule contains the genetic code
A) DNA B)mRNA C)tRNA D)rRNA
- c) Transcription occurs along the _____ direction in the template forming an mRNA in the _____ direction
A) 5' to 3'; 5' to 3' B) 5' to 3'; 3' to 5' C) 3' to 5'; 5' to 3' D) 3' to 5'; 3' to 5'
- d) Which mode of information transfer usually does not occur?
A) DNA to DNA B) DNA to RNA C) protein to RNA D) all occur in a working cell
- e) Which of the following is an example of the degeneracy of the genetic code?
A).Each amino acid is encoded by only one codon.
B).Some amino acids are encoded by multiple codons, reducing the effect of point mutations.
C).Each codon specifies more than one amino acid.
D) Degeneracy means that codons overlap with each other in translation.

Q2 Describe the form in which the cells store the information. What is a gene? **(2 marks)**

Q3 Describe the process by which nucleic acid sequence is translated into amino acid sequence **(2 marks)**

Q4 Describe the process of transcription. And list different enzymatic activities of the RNA polymerase **(2 marks)**

PartB (Answer any seven, 2 marks each)

Q1 Why ribosome are membrane less?

Q2 Give some examples of membrane bound organelles? Mention the significance of double membrane.

Q3 Why some cells have many nucleus?

Q4 What is the difference between nuclear matrix and nuclear lamina?

Q5 What are the two functions of the nuclear pore?

Q6 What is the location of cajal bodies, mention their function?

Q7 Where Nuclear Enriched Abundant Transcript 1 (NEAT1) are found inside cells?

Q8 Mention the function of promyelocytic leukemia nuclear bodies (PML bodies).

Q9 Why is Ran GTP concentrated in the nucleus?

Q10 What is Transitional ER(tER)?

Q11 How ribosome –nascent chain complex is directed to ER by Signal recognition particles and SRP receptor ?